Phase 1 — Two lanes off one database

Peterbilt Atlantic Digital Twin · Side Car Twin & Historical Archive · read-only, the live system is never written to

EVEglyphDesign

v3 · 9 sites PA01–PA09 · CAD · EN/FR

CDK Drive — dealer system of record

Nine Peterbilt Atlantic sites. Sales, F&I, service and fixed ops, parts, accounting, payroll. Phase 1 never writes to it. Work continues to be entered by hand exactly as it is today.

ACQUISITION — TWO INDEPENDENT LANES

Lane A · Backup set

Full restore into an environment the dealer controls. Exposes structure the API never will — table inventory, keys, code tables, application dictionary. Complete, trusted, lagging.

Delta refresh, seeded once

One full restore, then delta only. A refresh must cost in proportion to the volume of change, never to the size of the database. Native delta if CDK exposes one; our own utility if not. See page 2.

Lane B · Fortellis API

OAuth 2.0, Subscription-Id + Department-Id + Request-Id. CDK's own async pattern is Setup → Bulk → Delta. Current, partial, entitlement-bound.

The ledger is not reachable here

Chart of accounts, accounting schedules, department master, vendor master, warranty claim, purchase and receipt: all NONE on Fortellis. They can only come from Lane A or a file export.

LANDING AND CONFORMANCE — ONE POSTGRES

Postgres landing zone

Raw payloads staged as received, before any transform. Every row stamped with its lane, its source backup set or pull time, and its as-of marker. A bad transform is re-run against the same raw pull, never re-queried from CDK.

Data mart library — SAP-shape canonical core

Structure harvested from the application's own metadata, not hand-drawn. MARA / MARC / MARD / MBEW / MATDOC / MVKE / MARM / MFRPN and the FI/CO analogues, so a PACCAR SAP payload maps one to one and the dealer's model outlives any DMS vendor. Blanks in the API-derived structure are filled from Lane A. See page 3.

TWO MODELS, AND A FUNCTION THAT COMPARES THEM

Model A — straight off the backup

Runs from the restored copy to the 20 Group figures. Complete and self-contained. Answers: what does the closed record say.

Model B — from the API structure

Runs from the entitled API lane over the same conformed marts. Answers: what is the system saying right now.

Each lane produces the 20 Group set on its own. Neither depends on the other to be useful.

reconcile() — executable, on demand

Not a background process. Called when the answer is wanted, on a named scope and window. The API may overwrite the backup value within its authoritative scope, never silently; a later change showing variance is logged as an exception.

Reconciliation is the check that what happened in the system today matches what comes off the backup. It is not required for either to run.

Pour le bien-être du peuple.

Delta refresh — what we ask for, and what we build if the answer is no

The scalability rule, the questions for the CDK administrator, and the fallback utility

THE RULE THAT MAKES THIS SCALABLE

A refresh job's cost must be proportional to the volume of change, never to the size of the database. A run that scans beyond a declared proportion of an object halts and raises rather than proceeding quietly. Full-table reads happen on seed, or on an explicit and logged reseed, and at no other time.

What we ask CDK, precisely

Each answerable yes or no. Each answer unlocks a different strategy, or rules one out.

- Are incremental or differential backup sets available, or is the set full-only?
- Is there a transaction log backup chain, and is the log position or LSN exposed to us?
- Do objects carry a last-modified timestamp, row version, or monotonic sequence — and is it written on every write, including deletes?
- Are deletes hard or soft? A hard delete is invisible to a watermark pull.
- Is change data capture or change tracking enabled on any object?
- How is a specific backup set identified, what is the schedule, and what is the retention?
- Can a restore be taken more than once a day?
- Row counts and on-disk size of the ten largest objects.

Strategy A — native delta

If CDK exposes incrementals or a log chain. Seed once from a full restore, then apply the chain. The watermark is the backup set identity plus log position. A broken chain is detected on the next run and healed by reseeding only the affected objects.

Strategy B — our own utility

If the answer is no. Three tiers, chosen per object, with a cost guard on every run and a delta ledger recording rows scanned against rows changed.

Strategy B — three tiers, chosen per object

TIER	MECHANISM	WHEN IT APPLIES	COST BEHAVIOUR
Tier 1 Watermark	Bounded-range pull where the change column exceeds the last watermark, with a small overlap window for clock skew and late writes.	The object carries a reliable change marker written on every update.	Reads only what changed, plus the overlap. Cheapest tier. Blind to hard deletes.
Tier 2 Key + hash	Pull the primary key and a hash of the business columns only — a narrow projection, not full rows. Compare against the stored hash index.	No trustworthy change marker, or deletes must be detected.	Scans the key space but transfers almost nothing. Inserts, updates and deletes all fall out of the key and hash difference.
Tier 3 Key-range sweep	Rotate the object through key ranges so no single run touches the whole thing.	Large objects where even a key-space scan is too heavy in one run.	Bounded per run by construction. Full coverage is reached across a cycle, not in one pass.

Seed and reseed

Seed once: one full restore, with per-object row counts and checksums stored as the baseline. A reseed is a deliberate act — authorised by Tim Hawkins, logged, and justified against the cost of the delta runs it replaces. It is never the quiet default when something looks wrong.

What every run records

Backup set identity, the window, the objects touched, rows scanned against rows actually changed, checksums, duration, and outcome. The ratio of scanned to changed is the number that tells you whether this is still an affordable pipeline or has quietly become a nightly full extract wearing a delta's name.

The data mart library — structure discovered, not invented

Harvest the application's own metadata, shape it to the SAP model already in use at the other end of the string

1 · Harvest

From the backup restore — it exposes what an API never will.

- Object and table inventory
- Columns, data types, nullability
- Keys, relationships, indexes
- Code and lookup tables with their value domains
- Screen and report definitions where held
- The application dictionary itself

2 · Shape to the SAP model

Pattern reuse, not a claim that the dealership runs SAP. PACCAR does.

DEALERSHIP OBJECT	SAP PATTERN REUSED
Parts and unit master	Material master — MARA / MARC / MARD / MBEW / MVKE
Customers and vendors	Business-partner master data
Repair orders	Document header with line items
Parts movements	Movement document — MATDOC
Ledger postings	Accounting document, account and cost-object dimensions
Sites PA01–PA09	Plant and organisational hierarchy
Periods	Fiscal calendar and posting period

3 · Marts the library produces

MART	GRAIN
Parts inventory and movement	Part x site x movement
Service RO and labour	Repair order line and technician punch
Work in process	Open order, by state and age
Unit sales	Unit x deal
Financial performance	Account x department x period, absorption

Already held

21 objects and 443 confidence-tagged fields, with generated Postgres and Snowflake DDL and a preflight validator. Undocumented ledger-side objects are the first thing admin access is used to verify. Confidence tags travel with the column into the marts.

Filling the blanks

The API-derived structure will have gaps. They are filled from the backup lane. Every column in the conformed layer carries which lane supplied the value, its confidence tag, and its as-of marker — so a blank, a genuine null, and a field we are simply not entitled to see are three different things on the face of the record.

PRECEDENCE, IN THE OPERATOR'S WORDS

The API can overwrite what is in the backup-and-recovery lane, within its authoritative and entitled scope, and never silently: both values, the source and the timestamp are recorded on every overwrite. But if the next change arrives and there is a variance against what was previously reconciled, it is logged as an exception rather than quietly winning.

Exception classes

CLASS	TRIGGER	RESPONSE
Variance beyond tolerance	The two lanes disagree on a value by more than the declared tolerance for that field class.	Raised on the item, both values carried, neither lane declared correct by the model.
Row in one lane only	Present in the backup, absent from the API pull, or the reverse.	Checked against entitlement first — an entitlement gap is not a data defect.
Late-arriving change	A change lands against a period already reconciled.	Exception with the prior reconciliation run referenced, so the earlier answer stays explainable.
Schema or field drift	A column appears, disappears or changes type in either lane.	Load gated by the preflight validator; the library version is incremented.

Decision support and the work list — the twin proposes, the operator disposes

Two lists, drafts only, and a deliberate cost on pressing send

Conformed marts

Every figure originates in SQL over the marts. No numeric value is ever produced by the model.

Ranking, in SQL

Exposure, ageing, variance against the 20 Group set. Computed, auditable, re-runnable.

Azure-hosted open-source model, version pinned

Orders and explains. It does not adjudicate, and it does not produce numbers. Pinned so results stay comparable month to month.

TWO LISTS, DELIBERATELY NOT MERGED

By urgency

What will hurt if nothing is done today. Stock-outs against open work in process, ageing repair orders, exposure sitting on the accounting schedules, warranty drafts about to fall out of window.

By strategy

What compounds if it is started this week. Stocking quality against what actually sold, absorption drift by site, parts sitting dead at one of the nine sites and short at another, customers whose service interval has quietly lapsed.

Kept apart on purpose. Merge them and urgency eats strategy every single day, and the dealership never gets past today.

DISPATCH — DRAFT ONLY

The item

Site PA01–PA09, the observation, the figure with its as-of marker and which lane supplied it, why it ranked here, the suggested action, the contact, and the exposure in CAD. An unreconciled figure says so on its face.

Click, and a draft opens

Only on a click. Contact details resolved from master data, English or French from the contact's own held preference, never guessed. It opens unsent in the operator's own mail client. Nothing is written back to CDK — work is still entered by hand, as it is today.

The friction gate

An alien sits at the foot of every draft. Delete it, then send. A recipient who gets one with the alien still attached knows it was fired off unread. Configurable — Tim may take it off.

WHY THE FRICTION IS THE DESIGN

The failure mode of AI-assisted outreach is volume, not error. A send that costs nothing gets spent carelessly. Ease of dispatch is not a feature to be maximised here — the cost of sending is held deliberately above zero, and the joke is load-bearing, because it cannot be satisfied without actually reading the message.

Gates considered

GATE	WHAT IT ACTUALLY PREVENTS	VERDICT
Delete the alien	Sending without reading, and it is visible to the recipient when it happens.	Recommended to start. Rotate or retire it before it becomes furniture.
Skill-testing question	Inattention, briefly.	Learned and clicked through within a week.
State why this contact	Sending to the easy name rather than the right one.	Highest value, highest cost. Consider for high-exposure items only.
Daily send cap	Volume, bluntly.	Keep as a backstop, not as the gate.

What is measured

Not emails sent. Items acted on as a proportion of items shown, items ageing out untouched, exposure closed in CAD, and items that recur after being marked done. A rising send count against a flat outcome is a failure signal, not a success one.

The other side of the string — what does PACCAR want

Why a named counterpart is worth asking for now, while Phase 1 is still read-only and cheap to explain

Two choices in Phase 1 were made for reasons that only pay off if PACCAR is eventually in the room: the twin is read-only, and its core is SAP-shaped. PACCAR runs an SAP stack — ECC moving to S/4HANA, with TM, GTS, IBP, BFC and Concur around it. A dealer-side payload built in MARA / MATDOC / FI-CO shape therefore arrives in PACCAR's own modelling idiom, not as a DMS vendor's proprietary extract. That is unusual for a dealer to be holding, and it is the opener with a PACCAR representative.

The systems reveal the objectives

PACCAR has no public dealer API across most of its estate. But what a company chooses to build tells you what it wants. The column below is inference from evidence, and is marked as such.

Read from what PACCAR has actually built

PACCAR SYSTEM	WHAT BUILDING IT IMPLIES THEY WANT (INFERENCE)	WHAT A DEALER TWIN COULD OFFER TOWARD IT
Managed Dealer Inventory (MDI) Daily automated stock-order recommendation — Stock, Emergency, MKT, COF	Stocking quality controlled across the network, not left to nine separate judgement calls.	The loop closed: what MDI recommended, against what actually sold or sat, by site, with the movement history behind it.
Online Parts Counter (OPC) 545,000+ parts, dealer and fleet ordering	Order capture held in PACCAR's own channel rather than leaking to the aftermarket.	Visibility of where demand is being met off-channel, and why — availability, lead time, or price.
PRWS — Registration and Warranty System The one PACCAR system reachable through Fortellis	Clean warranty data captured at source, with fewer adjudication rounds.	Drafts raised from complete repair-order data, with the gaps that cause rejection visible before submission.
Syncron SLM with Returns SmartBlox Adopted 2022 for dealer-to-OEM returns	Returns friction taken out of the channel.	Return candidates identified from movement history rather than from a shelf someone happened to look at.
PACCAR Solutions / PSSM and SmartLINQ via Decisiv	The service event starting from the truck, not from a phone call.	The dealer-side half of that event recorded properly, so the case and the repair order tell the same story.

The win-wins, concretely

- Stock you did not know you had. PACCAR does not expose an inter-dealer transfer service. Nine sites are already holding parts that another site is short of, bought and paid for, sitting still. The twin creates that visibility outside ePortal. The dealer frees working capital; PACCAR gets a fill without shipping another unit from Renton.
- Stocking feedback into MDI. A recommendation engine with no return channel is guessing well rather than learning.
- Warranty data quality at source, through the one integration path that already exists.
- Returns identified from movement history rather than from memory.

THE LINE WE ARE NOT CROSSING

The boundary is kept on purpose. Read-only. No live writes. No bulk export of telematics — PACCAR's Truck Connectivity Services terms say plainly that PACCAR collects, uses and retains that data, and the twin is built to respect it rather than to test it. The discipline is what makes the relationship askable in the first place. Trust is the asset being built here.

The longer arc

Once the dealer-side record is clean, complete and shaped the way PACCAR already models the world, the dealer stops being a reporting obligation and starts being a feedback surface: real duty cycles, real failure patterns, real parts consumption in Atlantic Canadian winters, across nine sites. That is something PACCAR cannot buy. It is Phase 3 or later, it depends entirely on their appetite, and it is worth naming now only so that nothing built in Phase 1 forecloses it.

What we would ask for

None of it requires PACCAR to build anything new.

1. A named counterpart in dealer systems or PACCAR Parts.
2. Whether Peterbilt dealers report into NADA's ATD 20 Group composite, or a PACCAR-proprietary equivalent. Genuinely unresolved, and it shapes the surface.
3. EDI trading-partner setup for the group — 810, 850, 852, 855, 856, 861, 862. This is how the DMS vendors already do it.
4. Whether any PACCAR system supports service-account authentication for unattended read, rather than interactive dealer login only.
5. A conversation about telematics access, on terms PACCAR sets.

Pour le bien-être du peuple.

Investment perspective

A replica-first build that proves value on a copy of the data before any production system, budget, or rollout decision is asked for.

01
Zero production risk

The build works from the CDK backup, never the live dealer management system. There is nothing to negotiate about touching a running dealership because nothing live is touched. The replica is read and the performance report is extracted from it, while the working system carries on unaffected. Synthetic data stands in until Tim Hawkins's agreement is executed and Luke Weatherbie has 20-Group access.

02
The model, owned outright

The analysis layer runs on an Azure-hosted open-source model with a pinned version, chosen deliberately over a vendor API. Pinning removes silent model drift, so results stay comparable month to month rather than shifting under an opaque update. Cost is controlled and predictable, set by hosting rather than by per-token vendor exposure, which matters once ingestion and reporting run on a recurring basis.

03
The surface is the asset

The visible output is a single 20 Group performance dashboard, but the asset underneath it is the structured data built to support that view. Craig Allen's 16-system register is the substrate the schema is modelled against, so the surface reflects how the dealer group's systems actually fit together. Client-facing material is kept separate from the protected internal origin thesis behind the build.

04
Staged commitment

Phase 1 buys evidence cheaply: a working replica, an extracted performance report, and a dashboard, built on synthetic and mock data with no dependency on the live CDK system. Whether to write back into CDK, connect the TELUS call-history service account, or roll the pattern out more widely are separate decisions, made later, once Phase 1 has been seen and judged on its own evidence rather than promised in advance.

05
Room to replicate the pattern

If Phase 1 holds up, the same replica-first pattern can be extended in both directions: the trucks and the field on one end, and PACCAR logistics and production systems on the other. Nothing here commits to that extension now. It is a later, optional decision that the current architecture is deliberately shaped to make straightforward, without requiring the underlying approach to be rebuilt or renegotiated.

The commitment, plainly

A ceiling of $10,000, agreed

Phase 1 is scoped to the first investigation Tim Hawkins and EVEglyph Design agreed — a ceiling of ten thousand dollars. It is deliberately small. It buys evidence, not a rollout, and it is sized so that the decision to stop is as easy as the decision to carry on. Nothing in Phase 1 requires a production change, a licence, or a staff member to work differently.

What transfers at the end

Not a slide deck. The repository — the schema, the extract harness, the delta utility, the reconciliation function, the specifications, and the surface — forked to Hawkins on completion, under their own account. If the work is judged worthless, they still hold every artefact and can have someone else carry it on. That is the test we are willing to be held to.

This week, Phase 1 asks only for sign-off to build the replica and report on synthetic data.